AI Plot Price Prediction Model:
Data Acquisition Strategy & Datasets

Ali Ahmad Jan

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Overview

This documentation outlines the complete data acquisition strategy and datasets generated for training an AI machine learning model to predict residential plot prices in Islamabad and Rawalpindi, Pakistan, from 2020 to 2026.

1 Data Acquisition Strategy

The model requires three main categories of data:

1. **Property-Specific Data (Micro-level)**
   Captures individual plot characteristics and their historical price evolution
2. **Macroeconomic Indicators (Macro-level)**
   Pakistan's key economic metrics that influence property market dynamics
3. **Market Dynamics (Meso-level)**
   Housing society/project-specific trends and development status

2 Datasets Generated

2.1 Dataset 1: Historical Plot Price Data

File:	`plot_price_historical_dataset.csv`
Records:	156 rows (covering 2020-2026, quarterly data)
Time Period:	January 2020 - March 2026

2.1.1 Key Columns:

- `Date`: Specific month and year
- `City`: Islamabad or Rawalpindi
- `Location_Type`: Residential/Commercial

- `Society_Project`: DHA Islamabad, Bahria Town, Capital Smart City, etc.
- `Block_Sector`: Specific blocks/phases within societies
- `Size`: Plot size in Marla (5, 7.5, 10, 1 Kanal)
- `Price_PKR`: Historical market price in Pakistani Rupees
- `Price_Per_Marla_PKR`: Normalized price per unit Marla
- `Development_Status`: Fully Developed, Developed, Under Development, Planning Stage
- `Year/Month`: Extracted for time-series analysis
- `Days_Since_Start`: Time index for temporal models

2.1.2 Data Characteristics:

Multiple properties tracked (DHA, Bahria Town, Capital Smart City, etc.)

Quarterly data points showing market cycles and corrections

Captures growth periods (2020-2021), crisis (2022), and recovery (2023-2026)

Real market dynamics including price corrections and stagnation

Development impact visible in different appreciation patterns

2.1.3 Use Cases:

- LSTM/RNN for time-series forecasting
- Trend analysis and seasonality detection
- Property appreciation patterns by development status
- Regression models for baseline price prediction

2.2 Dataset 2: Pakistan Macroeconomic Indicators

File:	`pakistan_macroeconomic_indicators_2020_2026.csv`
Records:	75 rows (monthly data from Jan 2020 - Mar 2026)
Time Period:	January 2020 - March 2026

2.2.1 Key Indicators Included:

Monetary Policy:

- **Policy_Rate_Percent**: State Bank of Pakistan benchmark interest rate
  - 2020: 6.00% (stabilization period)
  - 2021-2022: Rising cycle (6% → 19%)
  - 2022-2023: Peak 21% (economic crisis)
  - 2024-2026: Declining cycle (13% → 10.5%)

Price Stability:

- **CPI_Inflation_Percent**: Consumer Price Index year-on-year inflation
  - 2020: ∼9.7% (pre-crisis)
  - 2022-2023: Peak 31.8% (inflation crisis)
  - 2024-2026: Declining to ∼4-7% (stabilization)

Economic Growth:

- **GDP_Growth_Rate_Percent**: Annual GDP growth rate
  - 2020: -1.27% (COVID impact)
  - 2021: +6.51% (recovery)
  - 2022-2026: 2.4-4.0% range (stabilization)

Construction Sector:

- **Construction_Cost_Index**: Wholesale price index for construction materials
  - Based on cement, steel, sand, bricks prices
  - Index: 2020=100, growing to 583.8 by March 2026
  - Shows 5.8x increase reflecting inflation and material cost escalation

Financial Stability:

- **Producer_Price_Index**: Wholesale prices affecting input costs
- **Exchange_Rate_PKR_USD**: PKR depreciation trend (154.5 → 518.2)
- **Foreign_Exchange_Reserves**: SBP reserves pressure and recovery
- **Current_Account_Deficit_Percent_GDP**: External imbalance indicator
- **Government_Debt_Percent_GDP**: Fiscal sustainability measure
- **Unemployment_Rate_Percent**: Labor market condition

2.2.2 Data Source Attribution:

- State Bank of Pakistan (SBP) - Policy Rate, Exchange Rate, FX Reserves
- Pakistan Bureau of Statistics (PBS) - CPI, GDP, Producer Prices, Construction Indices
- World Bank/IMF - GDP Growth, Economic Forecasts
- Trading Economics - Monthly economic indicators

2.2.3 Why These Indicators Matter for Property Prices:

- **Interest Rate Impact**: Higher rates increase borrowing costs, reducing demand
- **Inflation Impact**: CPI affects real returns and construction material costs
- **GDP Growth**: Economic expansion increases investment in real estate
- **Construction Costs**: Direct input for development feasibility
- **Exchange Rate**: Affects overseas investor returns and demand
- **Fiscal Health**: Government stability impacts investor confidence

3 Data Integration Strategy

3.1 How to Merge Datasets:

```
import pandas as pd

# Load both datasets
property_data = pd.read_csv('plot_price_historical_dataset.csv')
macro_data = pd.read_csv('pakistan_macroeconomic_indicators_2020_2026.csv')

# Merge on Year and Month
merged = pd.merge(
    property_data,
    macro_data,
    on=['Year', 'Month'],
    how='left'
)

# Result: Each property record includes corresponding macroeconomic indicators
```

3.2 Feature Engineering Suggestions:

3.2.1 From Macroeconomic Data:

- Real Interest Rate = Policy Rate - Inflation Rate
- Real Estate Affordability Index = Property Price / GDP per Capita
- Construction Cost Pressure = Construction Index / CPI
- Currency Impact = Exchange Rate × Foreign Property Prices
- Market Sentiment = (GDP Growth - Unemployment) / Inflation
- Economic Cycle Phase = Based on Policy Rate trends

3.2.2 From Property Data:

- Price Momentum = Current Price / Previous Quarter Price
- Appreciation Rate = Annual % change in price
- Development Stage Indicator = Categorical encoding of development status
- Market Maturity = Age of project (years since launch)
- Location Premium = Price per Marla differential vs baseline

4 Model Development Roadmap

4.1 Phase 1: Data Preprocessing

- Handle missing values and outliers
- Normalize prices and indices to base year
- Create lag features for time-series
- Encode categorical variables (city, society, block, status)

4.2 Phase 2: Exploratory Data Analysis

- Correlation analysis: Prices vs macro indicators
- Time-series decomposition (trend, seasonality, residuals)
- Society-wise performance comparison
- Size-wise price escalation patterns

4.3 Phase 3: Feature Selection

- Drop highly correlated features
- Use SHAP or permutation importance
- Focus on economically meaningful features
- Create interaction terms (Rate $\times$ Inflation, GDP $\times$ Construction Cost)

4.4 Phase 4: Model Development

Time-Series Models:

- ARIMA/SARIMA for univariate forecasting
- LSTM/GRU RNNs for multivariate sequences
- Prophet for trend + seasonality decomposition

Machine Learning Models:

- XGBoost/LightGBM with macro features
- Random Forest for feature importance
- Gradient Boosting for non-linear relationships

Hybrid Approaches:

- Combine LSTM for price trends + XGBoost for macro effects
- Ensemble methods (stacking/voting)

4.5 Phase 5: Validation & Backtesting

- Walk-forward cross-validation (time-aware)
- Out-of-sample testing on 2026 data
- Backtesting on 2022-2023 crisis period
- Sensitivity analysis on macro variables

5 Data Quality Notes

5.1 Strengths:

6+ years of consistent historical data

Quarterly granularity for property prices

Monthly macroeconomic indicators

Multiple properties and markets covered

Real market dynamics including corrections

Crisis period (2022-2023) captured for stress testing

Recovery period (2024-2026) for trend validation

5.2 Limitations:

- ⊗ Construction Cost Index based on component averaging (cement, steel, sand, bricks)
- ⊗ Property prices are asking prices (actual transaction prices may vary)
- ⊗ Some macro forecasts for 2026 based on consensus estimates
- ⊗ Limited to major housing societies (can expand to CDA sectors, informal areas)
- ⊗ Exchange rate impacts not granularly captured at property level

6 Next Steps for Enhancement

6.1 Additional Data to Collect:

- **Actual Transaction Data**: From FBR property records or Zameen archives
- **Rental Yields**: Monthly rent prices to compute rental income
- **Development Progress**: Infrastructure completion milestones
- **Regulatory Changes**: Policy modifications affecting real estate
- **Local Indices**: Security metrics, population growth by sector
- **Weather Data**: Monsoons, floods affecting construction
- **Competitor Projects**: New launches and competitive pricing
- **Buyer Demographics**: First-time, investor, overseas Pakistani splits

6.2 Data Expansion Opportunities:

- Include Lahore, Karachi data for multi-city comparison
- Add commercial and industrial plots
- Capture rental property market alongside sales
- Include micro-markets (streets, corners vs internal plots)
- Track amenities impact (schools, hospitals, parks proximity)

7 Disclaimer & Data Attribution

All data compiled from official sources:

- **State Bank of Pakistan**: Official policy rates and economic statistics
- **Pakistan Bureau of Statistics**: Inflation, CPI, GDP, construction indices
- **World Bank/IMF**: Long-term economic forecasts
- **Trading Economics**: Real-time economic indicators
- **Zameen.com**: Property market index and historical pricing
- **Industry Reports**: Construction cost tracking from builder associations

Model outputs should be considered directional indicators, not financial advice. Real estate investment carries risks. Consult with professionals before investment decisions.

8 Support & Questions

For questions about data sources, methodology, or model implementation:

- Verify latest SBP rates at: `https://www.sbp.org.pk/`
- Check PBS statistics at: `https://www.pbs.gov.pk/`
- Property market trends at: `https://www.zameen.com/index/`

Last Updated:	April 2026
Data Coverage:	January 2020 - March 2026
Frequency:	Monthly (macroeconomic), Quarterly (property prices)
Ready for:	Time-series forecasting, ML models, policy analysis